EXCERCISE 1 : SQL FUNDAMENTALS

1. SELECT *
   FROM employees_table;

id	first_name	last_name	department	Salary	hire-date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	CapeTown
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Mokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2. SELECT DISTINCT department
   FROM employees_table;

department
IT
HR
Finance
Marketing

3. SELECT first_name,
   last_name
   FROM employee_table
   ORDER BY salary DESC;

first name	Last name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

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4. SELECT id,
first_name,
last_name,
salary
FROM employees_table
ORDER BY salary
LIMIT 3;

id	first name	last name	Salary
1	Alice	Green	70 000
3	Cathy	Zulu	65 000
5	Eva	Naidoo	72 000

5. SELECT id,
first_name,
last_name,
department
FROM employees_table
WHERE department = 'IT';

id	first name	last name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT id,
first_name,
last_name,
department,
salary
FROM employees_table
WHERE department = 'Finance' AND salary > 60 000;

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id	first name	last name	department	Salary
3	Cathy	Zulu	Finance	65 000

7. SELECT id,
first_name,
last_name,
department
FROM employees_table
WHERE department = 'HR' OR department = 'marketing';

id	first name	last name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT id,
first_name,
last_name,
department
FROM employee_table
WHERE NOT department = 'IT';

We can also write the WHERE like this:
WHERE department != 'IT';

id	first name	last name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

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9. SELECT id,
first_name,
last_name,
department
FROM employees_table
WHERE department IN('IT', 'HR', 'Finance');

id	firstname	lastname	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id,
first_name,
last_name,
department,
Salary,
city
FROM employees_table
WHERE department='IT' AND Salary > 65 000
AND city='Johannesburg';

id	firstname	lastname	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg

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